

# ATM Interface System

A Java-based ATM simulation developed using Object-Oriented Programming (OOP) principles. The application allows users to perform basic banking operations such as depositing money, withdrawing money, and checking account balance through a menu-driven interface.

---

## ✦ Features

- Deposit money into a bank account
  - Withdraw money from a bank account
  - Check current account balance
  - Input validation for:
    - Invalid deposit amounts
    - Invalid withdrawal amounts
    - Insufficient balance
  - Menu-driven ATM interface
  - Exit option to terminate the program
- 

## ✂ Technologies Used

- Java
  - Object-Oriented Programming (OOP)
  - Scanner Class
  - Conditional Statements
  - Loops
  - Methods
- 

## 🏠 Project Structure

The project consists of two classes:

### 1. BankAccount

Responsible for managing account-related operations.

#### Attributes

- `balance`

#### Methods

- `deposit()`
  - `withdraw()`
  - `getBalance()` / `checkBalance()`
- 

## 2. ATM

Responsible for:

- Displaying ATM menu
  - Taking user input
  - Calling BankAccount methods
  - Managing program flow
- 

## Functionalities

### Deposit

- Accepts deposit amount from user
- Validates input
- Updates account balance

### Withdraw

- Accepts withdrawal amount
- Checks available balance
- Prevents overdraft transactions

### Balance Inquiry

- Displays current account balance

### Exit

- Terminates the ATM session
- 

## ► How to Run

### Step 1: Save the files

ATM.java  
BankAccount.java

## Step 2: Compile the program

```
javac ATM.java BankAccount.java
```

## Step 3: Run the program

```
java ATM
```

---

## OOP Concepts Implemented

### Classes and Objects

- BankAccount class
- ATM class
- Creation of BankAccount object inside ATM

### Encapsulation

- Account balance stored as a private data member
- Accessed and modified through methods

### Methods

- Deposit
- Withdraw
- Balance Check

### Input Validation

- Prevents invalid transactions
  - Ensures sufficient balance before withdrawal
- 

## Sample Menu

```
===== ATM MENU =====
```

1. Deposit
  2. Withdraw
  3. Check Balance
  4. Exit
- 

## Future Enhancements

- PIN Authentication

- Multiple User Accounts
- Transaction History
- Mini Statement Generation
- Fund Transfer Feature
- GUI Version using Java Swing or JavaFX
- Database Integration